

Project Title:

Second-hand Car Assistant

Group Member's Names:

Brianna Zaffina, Dylan Geraci

Proposed Problem:

When buying a used car, knowing how long it will last and how well it will perform is vital. Not everyone knows about cars, making it hard to see what a good deal would be vs. a deal that will end up more expensive than originally thought. As such, our project aims to provide those looking into buying second-hand cars with information regarding the probability of repairs.

Programming Language:

Python

Datasets:

<https://www.kaggle.com/code/satyaprakashshukl/used-car-price-prediction>

Algorithm/Approach:

We will use a Bayesian network to model dependencies on issues that specific variables might lead to. We will be using libraries such as pgmpy, pandas, and numpy.

Timeline:

April 12th - April 19th: Set up Python environment and install necessary libraries, connect CSV, and map the Bayesian nodes.

April 19th - April 26th: Clean dataset, create the training scripts, and test the models.

April 26th - May 3rd: Develop a UI and connect it to the backend model.

May 3rd - May 5th: Ensure there are no errors within the project and focus on developing and practicing the presentation.

May 5th - May 15th: Ensure the project is finished with no errors and develop the final report.

Roles & Responsibilities:

- Dylan: The AI Lead
- Designing the probabilistic model and the mathematical engine.
 - Model Architecture: Design the Directed Acyclic Graph (DAG) by deciding which variables (like Age or Brand) impact the "Failure" probability.
 - Bayesian Logic: Implement the pgmpy library to build the Bayesian Network and define the Conditional Probability Tables (CPTs).
 - Inference Engine: Write the backend functions that take user inputs and calculate the specific risk percentages.
 - Technical Validation: Test the model to ensure that a high-mileage car actually returns a higher risk score than a low-mileage one.
 - AI Documentation: Write the sections of the report explaining the Bayesian math and the reasoning behind the network's structure.
- Brianna: The Software & Data Engineer
- Focus: Managing the data pipeline and building the user-facing application.
 - Data Sourcing & Cleaning: Find the Kaggle dataset and use Pandas to clean it, filter out irrelevant columns, and "bin" numerical data (e.g., turning 150k miles into the "High" category).
 - Frontend Development: Build the Streamlit dashboard, including the sliders for user input and the visual gauges for the final recommendation.

- System Integration: Connect the frontend to the AI backend so that moving a slider instantly updates the "Risk Score" on the screen.
- DevOps & Deployment: Manage the GitHub repository and handle the technical setup to ensure the app runs smoothly for the class demo.
- Product Documentation: Create the user guide and the "Implementation" section of the final report, focusing on how the software was built.